

Finite Matrix Γ_3 -Contractions Without Γ_3 -Unitary Dilations

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Status: candidate full negative solution. There is an explicit sequence of commuting finite matrix triples $T^{(m)}$ such that every $T^{(m)}$ is a Γ_3 -contraction, but every sufficiently large $T^{(m)}$ has no Γ_3 -unitary dilation. The matrices have size $8m \times 8m$. This answers the matrix-dilation question in arXiv:1712.05707 negatively, already for $n = 3$.

1 Question and input from the literature

Let $\mathbb{G}_3$ be the symmetrized tridisc and $\Gamma_3 = \overline{\mathbb{G}_3}$. A commuting triple $T = (T_1, T_2, T_3)$ is a Γ_3 -contraction when Γ_3 is a spectral set for T . A Γ_3 -unitary is a commuting normal triple with joint spectrum in the distinguished boundary $b\Gamma_3$.

On PDF page 28 of Sourav Pal, *Rational dilation for operators associated with spectral interpolation and distinguished varieties*, arXiv:1712.05707, one finds:

Source question. *Does every Γ_n -contraction consisting of commuting matrices dilate to a Γ_n -unitary?*

An earlier paper of Pal, arXiv:1610.00425, constructs a Γ_3 -contraction on an infinite-dimensional Hilbert space whose adjoint has no Γ_3 -isometric, and hence no Γ_3 -unitary, dilation. We use the exact construction in Section 7 of that paper, but take finite sections. The new point is that the finite sections retain the scalar spectral-set inequality, while complete contractivity cannot persist along the whole sequence.

We also use the standard Arveson equivalence: since Γ_3 is polynomially convex, it is a complete spectral set for T if and only if T has a normal dilation with joint spectrum in its Shilov boundary $b\Gamma_3$.

2 The explicit finite matrix family

Put $E = \mathbb{C}^2$ and

$$X_1 = \begin{pmatrix} 0 & 1/4 \\ 0 & 0 \end{pmatrix}.$$

For $m \geq 1$, let $\mathcal{A}_m = E^m$. Define the truncated shift and the first-coordinate operator by

$$\begin{aligned} V_m(c_0, c_1, \dots, c_{m-1}) &= (0, c_0, \dots, c_{m-2}), \\ X_m(c_0, c_1, \dots, c_{m-1}) &= (X_1 c_0, 0, \dots, 0). \end{aligned}$$

On $\mathcal{H}_m = \mathcal{A}_m^4$, define

$$S_1^{(m)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & X_m & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad S_2^{(m)} = 0, \quad P^{(m)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & V_m & 0 & 0 \\ I & 0 & 0 & 0 \end{pmatrix}.$$

All entries in these displays are operators on $\mathcal{A}_m$. Consequently the three components are $8m \times 8m$ matrices. Set

$$U^{(m)} = (S_1^{(m)}, S_2^{(m)}, P^{(m)}), \quad T^{(m)} = (U^{(m)})^*.$$

Since $X_m^2 = 0$ and $X_m V_m = 0$, every product of two components of $U^{(m)}$ is zero. The same is true for $T^{(m)}$, so both are commuting triples.

For comparison, put $\mathcal{A} = \ell_2(\mathbb{N}_0; E)$, let V be the unilateral shift, and define

$$X(c_0, c_1, \dots) = (X_1 c_0, 0, \dots).$$

Using the same four-by-four block formulas gives the infinite-dimensional tuple $U = (S_1, 0, P)$ from arXiv:1610.00425. That paper proves

$$U \text{ is a } \Gamma_3\text{-contraction, but } U^* \text{ has no } \Gamma_3\text{-unitary dilation.} \quad (1)$$

3 The finite sections remain Γ_3 -contractions

Let $Q_m : \mathcal{A}^4 \rightarrow \mathcal{H}_m$ be the orthogonal projection onto the first m E -coordinates in every block. Componentwise,

$$U_j^{(m)} = Q_m U_j|_{\mathcal{H}_m} \quad (j = 1, 2, 3).$$

Compression does not usually preserve polynomial functional calculus. Here it does, for a special and decisive reason.

Lemma 1. *For every scalar polynomial f in three variables,*

$$f(U^{(m)}) = Q_m f(U)|_{\mathcal{H}_m}.$$

Consequently both $U^{(m)}$ and $T^{(m)}$ are Γ_3 -contractions.

Proof. Write

$$f(z_1, z_2, z_3) = a_0 + a_1 z_1 + a_2 z_2 + a_3 z_3 + R(z_1, z_2, z_3),$$

where every monomial of R has degree at least two. All products of two components vanish and the second component is zero, both for U and $U^{(m)}$. Hence

$$f(U) = a_0 I + a_1 S_1 + a_3 P, \quad f(U^{(m)}) = a_0 I + a_1 S_1^{(m)} + a_3 P^{(m)},$$

which gives the compression identity. By (1),

$$\|f(U^{(m)})\| \leq \|f(U)\| \leq \sup_{z \in \Gamma_3} |f(z)|.$$

Thus $U^{(m)}$ is a Γ_3 -contraction. If $f^\sharp(z) = \overline{f(\bar{z})}$, then $f((U^{(m)})^*) = f^\sharp(U^{(m)})^*$; the set Γ_3 is invariant under coordinatewise conjugation. The same inequality therefore holds for $T^{(m)} = (U^{(m)})^*$. $\square$

4 A complete-contractivity gap survives in finite dimension

Theorem 2. *There is $m_0 \geq 1$ such that, for every $m \geq m_0$, the commuting $8m \times 8m$ matrix triple $T^{(m)}$ is a Γ_3 -contraction but has no Γ_3 -unitary dilation.*

Proof. Regard $\mathcal{H}_m$ as a nested sequence of subspaces of $\mathcal{H} = \mathcal{A}^4$, extending each finite-section operator by zero on $\mathcal{H}_m^\perp$. The finite sections and their adjoints converge componentwise in the strong operator topology:

$$T_j^{(m)} \longrightarrow U_j^* \quad \text{strongly} \quad (j = 1, 2, 3).$$

They are uniformly bounded. It follows by expanding monomials that for every fixed matrix polynomial $F = [f_{ab}] \in M_d(\mathbb{C}[z_1, z_2, z_3])$,

$$F(T^{(m)}) \longrightarrow F(U^*) \quad \text{strongly on } \mathcal{H} \otimes \mathbb{C}^d. \quad (2)$$

By (1) and Arveson's theorem, Γ_3 is not a complete spectral set for U^* . Polynomial convexity of Γ_3 therefore supplies some d , a matrix polynomial F , and $\delta > 0$ such that

$$\|F(U^*)\| > C + 3\delta, \quad C := \sup_{z \in \Gamma_3} \|F(z)\|. \quad (3)$$

Choose a unit vector ξ with finite coordinate support such that $\|F(U^*)\xi\| > C + 2\delta$. This is possible because finite-support vectors are dense; one first chooses an approximate norming vector in (3) and then approximates it. For all sufficiently large m , $\xi \in \mathcal{H}_m \otimes \mathbb{C}^d$, and (2) gives

$$\|F(T^{(m)})\| \geq \|F(T^{(m)})\xi\| > C + \delta.$$

Thus Γ_3 is not a complete spectral set for $T^{(m)}$ for any sufficiently large m . A Γ_3 -unitary dilation would imply the opposite matrix polynomial inequality by compression of F of the normal dilation. Therefore no such dilation exists. Lemma 1 already shows that each $T^{(m)}$ remains a scalar Γ_3 -contraction. $\square$

The theorem gives a full negative answer to the source question. Its only non-effective feature is the value of m_0 : the supporting paper detects non-dilation through a structural fundamental-operator obstruction rather than displaying a particular matrix polynomial F . Once any witness F is extracted, the proof above makes the finite threshold effective.

5 Audit and novelty boundary

The accompanying verifier constructs the matrices for $1 \leq m \leq 8$ and checks their dimensions, contraction bounds, degree-two-zero relations, commutativity of the adjoints, and exact blockwise compression compatibility. These computations audit the explicit algebraic family; the decisive non-dilation step is the theorem-level complete-contractivity argument above.

A bounded literature search on 13 August 2026 covered the exact question, finite-dimensional and matrix Γ_3 -dilation, symmetrized-tridisc rational dilation, and finite-section variants. It found the known infinite-dimensional counterexample and later conditional dilation/model papers, but no finite-matrix answer or the argument above.

References

- [1] S. Pal, *Rational dilation for operators associated with spectral interpolation and distinguished varieties*, arXiv:1712.05707.
- [2] S. Pal, *Rational dilation on the symmetrized tridisc: failure, success and unknown*, arXiv:1610.00425.
- [3] W. Arveson, *Subalgebras of C^* -algebras. II*, Acta Mathematica 128 (1972), 271–308.